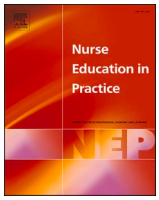
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# Contextualizing the digital competence framework for nursing education: Findings from a Delphi

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ABSTRACT

*Aim:* To identify key digital competencies for higher education nursing faculty, explore challenges in digital integration, and provide recommendations for systematic faculty development.
*Background:* Digital competencies are essential for nursing faculty to deliver high-quality education in an increasingly digitalised healthcare context. Despite their importance, faculty members often encounter challenges in integrating digital technologies into teaching, clinical education, and simulation, highlighting the need for structured and context-specific faculty development.
*Design:* A mixed-methods, consensus-based Delphi study.
*Methods:* A four-round Delphi study was conducted with nursing faculty from five European countries involved in the Erasmus + Digital Education in Nursing project. Twelve experts participated in the first round, increasing to 45 in the final round. Data collection combined qualitative feedback and quantitative ratings of digital competencies derived from the European Digital Competence Framework for Educators (DigCompEdu). Iterative rounds enabled refinement and consensus-building to contextualise the framework for nursing education.
*Results:* Six overarching competence domains were identified: Professional Engagement, Digital Resources, Teaching and Learning, Assessment, Empowering Learners, and Facilitating Learners' Digital Competencies. Participants reported well-developed basic digital skills related to literacy, communication, and safe technology use. In contrast, advanced competencies such as digital content creation, copyright awareness, programming, and problem-solving were less developed. Digital competencies were perceived as most applicable in theoretical teaching, with limited integration in clinical and laboratory-based education.
*Conclusion:* Nursing faculty view digital competencies as essential for learner-centred teaching, reflection, and professional development, but gaps persist in practice-based education. Stronger institutional support, faculty development, and adapted frameworks are needed to improve integration.

## 1. Introduction

The rapid digitalisation of higher education and healthcare has transformed how knowledge is taught, communicated and applied in practice. In this context, digital competencies have become essential for higher education teachers, particularly in nursing, where teaching must respond to both pedagogical demands and the digital transformation of clinical environments. Digital competencies of higher education teachers are crucial for high-quality and inclusive education in a digitally transformed society. Their importance is particularly pronounced in nursing, where education occurs in a highly regulated and rapidly evolving healthcare environment (Jobst et al., 2022; Vičič Krabonja,

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